PANJAB UNIVERSITY, CHANDIGARH

Outlines of tests, syllabi and courses of reading in the subject of Geography for M.Phil course (Semester system) for the session 2013-14.

The details of the courses and the dissertation semester wise shall be as under

FIRST SEMESTER

A. ***Compulsory Courses***

I. Recent Issues in Geographic Thought (4 Credits: 100 marks)
II. Research Methodology in Geography (4 Credits: 100 marks)

B. III. ***Optional Courses***

Each student shall be required to do one of the following optional courses:
(4 Credits: 100 marks each)

(i) An Advanced Course in Agricultural Geography
(ii) An Advanced Course in Fluvial Geomorphology
(iii) An Advanced Course in Political Geography
(iv) An Advanced Course in Population Geography
(v) An Advanced Course in Geography of Rural Settlements
(vi) An Advanced Course in Urban Geography
(vii) An Advanced Course in Urban Research Methods
(viii) An Advanced Course in Geography and Ecosystems

C. Dissertation (6 Credits: 150 marks)

SECOND SEMESTER

D. ***Compulsory Courses:***

IV. Geography of Regional Development (4 Credits: 100 marks)
V. Digital Cartography, Map Design and Remote Sensing (2 Credits: 50 marks)

E. ***Dissertation*** **(continued from First Semester)**

FIRST SEMESTER

Compulsory Courses
Course I: Recent Issues in Geographic Thought

Max. Marks	: 100
Terminal Examination	: 80
Internal Assessment	: 20
Time	: 3 Hours

Objectives: To familiarize the students about the latest philosophical and methodological issues in Geography. To educate them about the relevance of these new trends and their spatial contexts.

UNIT I

(i) History of Geography: Methodological issues and approaches
(ii) Theory and Geography

UNIT II

(iii) Quantification and the Development of Theory in Human Geography
(iv) Revival and Restructuring of Ecological Studies in Geography
(v) Revival and Restructuring of Regional Geography

UNIT III

(vi) General Systems Theory and Geography
(vii) Post- modernism and Geography

UNIT IV

(viii) Feminism and gender issues in Geography
(ix) Geography's research frontiers

Note:

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. The student shall attempt any 10 parts in about 25-30 words each. Each part shall carry 2 marks (total 20 marks).

2. A total of ***eight*** questions will be set out of the whole syllabus, at least *two* from each unit. The candidates will attempt ***four*** questions selecting one from each unit. These will be in addition to the compulsory question at serial number I and each question will carry 15 marks.

3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.

Pedagogy:

The students are exposed to latest research publications from journals like ANNALS Progress in Human Geography, Progress in Physical Geography etc so that they are aware of new research. They are encouraged to participate in seminars and special lectures are arranged on topics related to recent philosophies in Geography.

***Books Recommended*:**

Essential Readings

1. Abler, R.., Adams, J.S. and Gouid, P.R. (1971): *Spatial Organisation: The Geographer's View of the World*. Engiewood Cliffs, New Jersey, Prentice Hall.

2. Chorley, R.J. (ed.) (1973): Directions in Geography, Methuen & Co. Ltd.

3. Gatrell, A.C. (1983): *Distance and Space: A Geographic Perspective*, London, Oxford University Press.

4. Gregory, D. and R. Walford (eds) (1989); Horizons in Human Geography, London, Macmillan.

5. Johnson, R.J. (1985): *The Future of Geography*, London and New York, Methuen.

6. Stoddart, D.R. (ed) (1977): *Geography, Ideology and Social Concern*, Basil Blackwell.

7. Momscn. Janet H. and Townsend, Janet (eds): *Geography of Gender in the Third World*, Hutchinson, London, 1987.

8. *Regional Geography: Current Developments and Future Prospects*, edited by R.J. Johnston, J. Hauer and G.A. Hoekveid.

Further Readings

1. Board, Christopher, Chorley, Richard J., Haggett, Peler, Stoddart, D.R. (eds.): *Progress in Geography, International Reviews of Current Research,* Vols. I to VI11, London, Edward Arnold.

2. Dohrs, F.E. and Somers, L.M. (eds); *Introduction to Geography: Selective Readings*, Thomas Y. Crowell Co. New York, 1967.

3. Gregory, D. (I978): *Ideology, Science and Human Geography.*

4. Soja Edward, W. *Postmodern Geographies: The Reassertion of Space in Critical Social Theory*, Verso, London, 1990.

5. Women and Geography, Study Group of the 1EG, *Geography and Gender*, Hutchinson, London, 1984.

6. Daniels. Bradshaw and Others: *Human Geography: Issues for the 21st Century*, Pearson Education, Delhi, 2003.

Course II: Research Methodology in Geography

Max. Marks	: 100
Terminal Examination	: 80
Internal Assessment	: 20
Time	: 3 Hours

Objective: To train the students about various aspects of research methodologies. To educate them about the research writing and the new tools of data processing and analysis.

UNIT I

(i) Concept and basics of research methodology; the research process; parameters of Geographic research.

UNIT II

(ii) Discovery of research questions and formulation of the topic; construction of research design; research proposal

(iii) Data management- collection, processing, representation, analysis and interpretation of data

UNIT III

(iv) Use of cartography, quantitative analysis and G.I.S. in geographic research

UNIT IV

(v) Writing of the dissertation/ thesis

(vi) Defence of the thesis at viva voce

Note:

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. The student shall attempt any 10 parts in about 25-30 words each. Each part shall carry 2 marks (total 20 marks).

2. A total of ***eight*** questions will be set out of the whole syllabus, at least *two* from each unit. The candidates will attempt ***four*** questions selecting one from each unit. These will be in addition to the compulsory question at serial number I and each question will carry 15 marks.

3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.

Pedagogy:

Small exercise on report writing, synopsis preparation is given to students. Training on latest tools are arranged for these students.

Books Recommended:

Essential Readings:

1. C.R. Kothari: *Research Methodology'*, Wiley Eastern Limited, New Delhi, 1988.

2. H.N. Misra and Vijay P. Singh: *Research Methodology in Geography*, Rawat Publications, Jaipur 1998.

3. Peter Haggett and others: *Locational Analysis in Human Geography*. Arnold, London, 1977.

4. R.P. Misra*: Research Methodology*, Concept Publishing Company, New Delhi, 1989.

Further Readings:

1. David Harvey: *Explanations in Geography*, Arnold, London, 1969.

2. Ralph Berry: The Research Project: How to Write it, Routledge, London, 1990.

3. Thomas, S. Kuhn: *The Structure of Scientific Revolutions*. University of Chicago Press. Chicago, 1970.

4. Montello, Danial R. and Paul C. Sutton : An Introduction to Scientific Research Methods in Geography, Sage Publications, London, 2006.

B. III. Optional Courses

Opt. (i): An Advanced Course in Agricultural Geography

Max. Marks	: 100
Terminal Examination	: 80
Internal Assessment	: 20
Time	: 3 Hours

Objective: The main objective of this course is to familiarize the students with advancement in content and methodology of Agricultural Geography. The course also aims to provide an insight into Indian Agriculture with special reference to its regional divisions.

UNIT I

1. Dynamics of agricultural geography
2. Diffusion of agricultural innovations and agricultural transformation

UNIT II

3. Agricultural co-operation, contract farming and agri-business
4. Agriculture in the rural-urban fringe

UNIT III

5. Research methodology in agricultural geography
6. Agricultural Regions of India

UNIT IV

7. Sources of data and problems
8. Explanations and interrelationships in agricultural geography (use of correlation and regression analysis)

Note:

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. The student shall attempt any 10 parts in about 25-30 words each. Each part shall carry 2 marks (total 20 marks).

2. A total of ***eight*** questions will be set out of the whole syllabus, at least *two* from each unit. The candidates will attempt ***four*** questions selecting one from each unit. These will be in addition to the compulsory question at serial number I and each question will carry 15 marks.

3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.

Pedagogy:

Students should be encouraged to debate and discuss the various issues as well as developments taking place in the field of agricultural geography. Field visits should be organized to familiarize them with the ground reality of agricultural landscape.

Books Recommended:

Essential Readings:

1. Husain M, *Systematic Agricultural Geography*, Inter India Publications, Delhi, 1996.

2. IIbery, B.W: *Agricultural Geography*, Oxford University Press, Oxford, 1985.

3. Morgan, W.B. and Munton, RJ.C*: Agricultural Geography*, Methuen, London, 1971.

4. Singh. J. : *Agricultural Geography*, Oxford, New Delhi, 2003.

5. Taylor, P.J.: *Quantitative Methods in Geography*: *An Introduction to Spatial Analysis* Houghlon, Mifflin, Boston, 1977.

6. Tarrant, John -.*Agricultural Geography*, David and Charleys, Newton Abbot, Devon, 1974.

Further Readings:

1. Abler, Ronald. Adams, John, S. Gould Peter: *Spatial Organisation*: *The Geographer's View of the World*, Prentice Hall, N.J., 1971.

2. Hammond.R and Mc Cullagh. P.S.: *Quantitative Techniques in Geography*: *An Introduction.* Clareden Press, Oxford, 1977.

3. Hagget. P. and Chorley, R.J. (eds): *Socio-Economic Models* in Geography, Methuen & Co., London, 1976.

Opt. (ii): An Advanced Course in Fluvial Geomorphology

Max. Marks	: 100
Terminal Examination	: 80
Internal Assessment	: 20
Time	: 3 Hours

Objectives:

- This course represents the interface between physical aspects of Geography and Geology, Oceanography, Glaciology etc. The course aims to sensitise the student to this interface.
- The course aims to familiarize the student with the conceptual framework for understanding the existing geomorphological landscapes and related processes.

The course is designed to provide the student with a theoretical and empirical framework for understanding landscape evolution and the characteristics of individual types of geomorphic landscapes

UNIT I

(i) Fluvial System and Geomorphic Concepts
(ii) Drainage System Variables and Rates of Change

UNIT II

(iii) Transportation of Sediment Load
(iv) Fluvial Processes: Erosion and Deposition

UNIT III

(v) Dynamic Equilibrium and the Steady State
(vi) Channel Morphology and Channel Pattern

UNIT IV

(vii) Drainage Patterns and their Evolution
(viii) Analysis of Drainage Basin as a Geomorphic Unit

Note:

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. The student shall attempt any 10 parts in about 25-30 words each. Each part shall carry 2 marks (total 20 marks).

2. A total of ***eight*** questions will be set out of the whole syllabus, at least *two* from each unit. The candidates will attempt ***four*** questions selecting one from each unit. These will be in addition to the compulsory question at serial number I and each question will carry 15 marks.

3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.

Pedagogy:

The study of this paper needs adequate understanding of geomorphic forms and processes. It can be achieved through suitable use of audio-visual aids, photographs, maps, other forms of illustrations and, depending upon feasibility, field visits.

Books Recommended:

Essential Readings:

1. Davis, W.M.; *Geographical Essays*, New York, Dover, 1954.

2. Geogovy and Welling, D.W.: *Drainage Basin Form and Process*, New York, J.Willey, 1973.

3. Lepold, L.B. et. al: *Fluvial Processes in Geomorpholgy*, San Francisco, W.H. Freeman, 1964.

4. Morisawa, M.: *Fluvial Geomorpholog*y, Binchamton, S.V. of N.Y., 1973.

Further Readings:

1. Morisawa, M.: *Streams*, New York, McGraw Hill.

2. Pussell, R.J.*: River Plains and Sea Coasts*, Berkeley, U.C. Press, 1967.

3. Schumm, S.A.: Fluvial System, London, J. Willcy, 1977.

Opt. (iii): An Advanced Course in Political Geography

Max. Marks	: 100
Terminal Examination	: 80
Internal Assessment	: 20
Time	: 3 Hours

Objectives:

- The course is designed to acquaint the student with the conceptual framework for the study of geographical bases for political phenomena.
- To create awareness about the role of geographical factors in influencing political character of individual countries/regions.

Unit I

(a) Recent Advances in Political Geography:
  (i) Systems approach
  (ii) Behavioural studies

Unit II

(b) Values, altitudes and images:
  (i) A framework for decision-making
  (ii) Values and ideology
  (iii) Attitudes- political socialisation, attitude formation, spatial awareness
  (iv) Images of the political world- the process of image formation

Unit III

(c) Voting and Elections:
  (i) Methods of participation
  (ii) Social correlates of voting
  (iii) Social cleavages and Party System
  (iv) Influence of local environment on voting behaviour
  (v) Territorial Bases in Electoral Systems

Unit IV

(d) Politics and the Environment:
  (i) Public Opinion and the environment
  (ii) Pressure group and group activity
  (iii) Institutional Response
  (iv) Land use conflict.

Note:

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. The student shall attempt any 10 parts in about 25-30 words each. Each part shall carry 2 marks (total 20 marks).

2. A total of ***eight*** questions will be set out of the whole syllabus, at least *two* from each unit. The candidates will attempt ***four*** questions selecting one from each unit. These will be in addition to the compulsory question at serial number I and each question will carry 15 marks.

3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.

Books Recommended:

Essential Readings

1. Carlson, Lucille: *Geography and World Politics*, Prentice Hall. N.J., 1958.

2. Cohen, S.B.: *Geography and Politics in a Divided World*, Methuen & Co., London, 1964.

3. De Blij, 1 l.J.: Systematic Political Geography, John Wiley & Sons, Inc New York, 1973.

4. Fisher, Charles, A.: *Essays in Political Geography*, Methuen & Co., London, 1968.

5. Glassner, M.T. and D. Blij, H.J.*: Systematic Political Geography*, John Wiley & Sons, New York, 1980.

6. Pounds, N.J.G.: *Political Geography*, McGraw Hill, N.Y., 1968.

Further Readings

1. Cantori, Louis, J. Sleven, I. Spiesel : *The International Politics of Regions*: A Comparative Approach. Prentice Hall Inc, Engiewood. Cliffs, N.J. 1970.

2. Dikshit, R.D.*: Political Geography*: *A Contemporary Perspective*, Tata McGraw Hill Publishing Co. New Delhi, 1982.

3. East.W. Gordon and J. R. V. Prescott: Our Fragmented World: An Introduction to Political Geography. Crane Russak & Co., Inc New York. 1975.

4. Kaspcrson, Rogcrs, E.(ed.): The Structure of Political Geography, University of London Press, London, 1970.

5. Muir,R.: *Modern Political Geography*, Macmillcn Press, London, 1975.

6. Prescolt, J.R.V.: *The Geography of State Policies*, Aldine Publishing Co, Chicago, 1969.

7. Prcscott, J.R.V.: *Political Geography*, Methuen & Co. London, 1972.

8. Richard Muirand Rohan Paddison: *Politics, Geography amd Behavior*, Methuen, London, 1981.

9. M.A. Busteed (eds) : *Developments in Political Geography*, Academic Press, London, 1983.

Pedagogy:

The student should be encouraged to engage in classroom discussions on the geographical aspects of political problems at the national and international levels.

Opt. (iv): An Advanced Course in Population Geography

Max. Marks	: 100
Terminal Examination	: 80
Internal Assessment	: 20
Time	: 3 Hours

Objectives: The aim of this paper is to apprise the prospective researchers with latest advancements in the understanding of population issues in the context of regional synthesis and regional development in ever changing space-time continuum**.**

UNIT I

(i) Approaches to the study of Population Geography with a special reference to recent trends

UNIT II

(ii) Demographic Bases of Population: crude birth rate. fertility ratio and reproduction or degree of replacement; crude death rate. age and sex specific mortality rates, infant mortality rates/and maternal mortality rate, survival rate; longevity of life; primary, secondary and tertiary sex ratio.

UNIT III

(iii) Demographic Transition: concept, space-time diffusion, and spatial dimension of the demographic transition in India

(iv) Theories of migration: Ravenstein, W.J.Reilly, George K. Zipf. S.A.Stouffer, Hagerstrand, J.Wolpert, A.K Mabogunjc and Lee

UNIT IV

(v) A detailed study of population geography of India with reference to growth of population, urbanization, migration, economic composition, age-sex composition.

(vi) Population growth and regional development

Note:

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. The student shall attempt any 10 parts in about 25-30 words each. Each part shall carry 2 marks (total 20 marks).

2. A total of ***eight*** questions will be set out of the whole syllabus, at least *two* from each unit. The candidates will attempt ***four*** questions selecting one from each unit. These will be in addition to the compulsory question at serial number I and each question will carry 15 marks.

3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.

Pedagogy:
Good quality latest maps, classroom discussions, authentic audio-visual aids would be used to understand complex dimensions of population.

Books Recommended:

Essential Readings:

1. Chandna,R.C: Regional Planning and Development, Kalyani Publisher, New Delhi; 2008.

2. Chandna,R.C: A Geography of Population, 8th edition, Kalyani Publishers, New Delhi; 2009.

3. Chandna, R.C: Jansankhya Bhoogal, 7th edition, Kalyani Publishers, New Delgi, 2010.

4. Clarke,J.L: Population Geography, Pergamon Press Oxford, 1972 (2nd edition).

5. Dcmko,GJ. & et. al: *Population Geography: A Reader*, McGraw Hill Co., New York. 1970.

6. Hauscr,P.M. & Duncan.O.D.: *The Study of Population*. University Press Chicago, 1959.

7. Hornby,F.William and Jones,M.: *An Introduction to Population Geography*, Cambridge University Press, Cambridge. 1987.

8. Jones,U.K.: *A Population Geography*, Harper & Row Publishers, London, 1981.

9. Kayastha,S.L.: *Geography of Population*: Selected Essays, Rawat Publications, 1998.

10. Weeks, John, R.: *Population: An Introduction to Concepts and issues*, Wadsworth/ Thomson Leaning Belmout, LA, USA, 2002.

11. Zelinsky,W and et. al: *Geography and a Crowding World*, University Press, Oxford, 1970.

Further Readings:

1. Barclay,G.W.: Techniques of Population Analysis, John Wiley & Sons, New York, 1958.

2. Berelson,B. (ed.): Population Policy in Developed Countries, McGraw Hill Book Co., 1974.

3. Bose,Ashish: Population Profile and Religion in India, B.R.PubIications, Delhi. 1997.

4. Crook. Nigel (ed.): Principles of Population and Development, Oxford University Press, 1997.

5. Ford.T.R. and Delong.F.F: *Social Demography*. Prentice Hall, Inc., London. 1970.

6. Knowled, R.& Wareing,J: Economic and Social Geography: Made simple Books, Rupa & Co, New Delhi, 2007.

7. PohIman,E.: *Population: A Clash of Prophets*, New American Library, New York, 1973.

8. Premi,Mahendra K: India's Changing Population Profile National Book Trust India, New Delhi, 2009.

9. Robinson.W,C.: *Population and Development Planning*, The Population Council, New York, 1976.

10. Sundaram.K.V. and Nangia, Sudcsh (cds.): *Population Geography*, Heritage Publishers. New Delhi, 1985.

11. Woods, Robert: *Theoretical Population Geography*, Longman Inc., New York, 1982.

Opt. (v): An Advanced Course in Geography of Rural Settlements

Max. Marks	: 100
Terminal Examination	: 80
Internal Assessment	: 20
Time	: 3 Hours

Objectives:

- To acquaint the student with the conceptual foundations of this branch.
- To enable the student to understand the evolutionary, morphological and, functional attributes of rural settlements.
- To understand the features of different types of rural settlements in India and the transformation of the rural landscape.

UNIT I

(i) Recent approaches to the study of Rural Settlement Geography:
(a) Systems
(b) Culture- Ecological
(c) Culture- Historical
(d) Environmental perception

UNIT II

(ii) Settlement Models: Models of settlement diffusion: Bylund, Chisholm, and Grossaon.
(iii) Processes of Settlement Formation, Agglomeration.

UNIT III

(iv) Two Fundamental Components of Rural Settlements:

(a) Rural House Types and Regional Rural Cultures: Alternative theories of house form based on socio-cultural factors, climatic modifications, construction materials, and technology; evolution, diffusion and distributional patterns of different house types in India.

(b) Descriptive Analysis of Field Patterns; Field Systems and Field Patterns, classification, genesis and empirical attributes.

UNIT IV

(v) Transformation of Rural Settlements: Changes generated by modernization and governmental planning; Changes in Field Patterns, house types, and in the type, form, and patterns of rural settlements.

Note:

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. The student shall attempt any 10 parts in about 25-30 words each. Each part shall carry 2 marks (total 20 marks).

2. A total of ***eight*** questions will be set out of the whole syllabus, at least *two* from each unit. The candidates will attempt ***four*** questions selecting one from each unit. These will be in addition to the compulsory question at serial number I and each question will carry 15 marks.

3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.

Pedagogy:

The course should be conducted with the help of adequate illustrations using audio-visual aids and, depending upon feasibility, field visits.

Books Recommended:

Essential Readings:

1. Chisholm, M.: Rural Settlement and Land Use, London, Hutchinson University Library, 1962.

2. Haggett P. : Locational Analysis in Human Geography, London, Edwin Arnold, 1969.

3. Hudson, R.S.: A Geography of Settlements. London, MacDonald & Sons, 1970.

4. Mitra,A.: Report on House Types and Village Settlement Patterns in India, Delhi, Manager of Publications, 1966.

5. Mukerji.R.K.: Man and His Habitations. Bombay, Popular Parkashan, 1968.

6. Rapoport A. Amos: House Form and Culture, New Jersey, Prentice Hall, 1969.

Further Readings:

1. Ambrose, P.: Settlement Patterns, London, Longmans, 1970.

2. Forde, C.D.: Habitat. Economy and Society, London, Methucn & Co.. 1970.

3. Mac Gauch. M.E.: A Geography of Population and Settlement. Dubugue, W.C. Brown Company, 1970.

4. Maguirc. P.: From Tree Dwelling to New Towns. Bombay, Orient and Longmans, 1968.

5. Money. D.C.: Patterns of Settlement, London, Evans Brothers, 1972.

Opt. (vi): An Advanced Course in Urban Geography

Max. Marks	: 100
Terminal Examination	: 80
Internal Assessment	: 20
Time	: 3 Hours

Objectives:

- To familiarise the students with the theoretical foundations and recent trends in this branch of Geography.
- To provide an understanding of evolutionary, morphological and, functional attributes of urban places at different scales.
- To sensitize the students about contemporary urban problems.

UNIT I

(i) The Study of Urban Geography:
Place of Urban Geography in the context of Geography; History of the development of Urban Geography.

(ii) Recent trends in Urban Geography; Problems in the study of Urban Geography.

UNIT II

(iii) Urban Place as a Region; Analysis of Townscape: components of the townscape; physical and cultural influences on the components; evaluation of different components for townscape analysis.

UNIT III

(iv) Study of the townscapes of India with special reference to:
(a) Hill Resort Towns
(b) Cantonment Towns, and
(c) Walled Towns.

(v) Urban Landuse: detailed study of the models of landuse; landuse patterns in accidental and oriental towns.

UNIT IV

(vi) Urban Places in a Region: Spatial Organisation System of Urban Places; aspects to be studied: unit of study, level of generalization, methods of study.
Theoretical bases and generalisations on urban systems

Note:

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. The student shall attempt any 10 parts in about 25-30 words each. Each part shall carry 2 marks (total 20 marks).

2. A total of ***eight*** questions will be set out of the whole syllabus, at least *two* from each unit. The candidates will attempt ***four*** questions selecting one from each unit. These will be in addition to the compulsory question at serial number I and each question will carry 15 marks.

3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.

Books Recommended:

Essential Readings:

1. Berry. B.J.L. and F.E. Horton (eds) (1970). *Geographic Perspectives on Urban Systems*. Prentice Hall, Englewood Cliff, New Jersey.

2. Carler, H. (1972). *The Study of Urban Geography*, Edward Arnold, London.

3. Smailes. A.E. (1966). *Geography of Towns.* Hutchinson, University Press, London.

Further Readings:

1. Bird, James (1977). *Centrality and Cities*, Routledge & Kegan Paul, London.

2. Johnson. J.H. (1968). *Urban Geography: An Introductory Analysis*, Pergamon Press, New York.

Pedagogy:

The course should be backed up with extensive use of examples from Indian urban places with the help audio-visual aids and, depending upon feasibility, field trips.

Opt. (vii): An Advanced Course in Urban Research Methods

Max. Marks : 100
Terminal Examination : 80
Internal Assessment : 20
Time : 3 Hours

Objectives: The course aims at exposing the students to subtle conceptual issues and advanced methodological tools and techniques required to study urbanization and the processes of urbanization from a geographical perspective. The main focus is placed on tools and techniques that are applicable to process of urbanization in the developing course.

Unit I

(i) The process of urbanisation- approaches to the study of urbanisation.

(ii) Defining and delimiting urban areas.

Unit II

(iii) City as a social system, with special reference to social area analysis.

(iv) City as an economic system, with special reference to models of city growth and basic/ non basic ratio.

Unit III

(v) Urban- rural interaction and methods of delimiting umland.

(vi) Urban- rural differentials.

Unit IV

(vii) Analysis of central place systems with special reference to determination of hierarchy.

Note:

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. The student shall attempt any 10 parts in about 25-30 words each. Each part shall carry 2 marks (total 20 marks).

2. A total of ***eight*** questions will be set out of the whole syllabus, at least *two* from each unit. The candidates will attempt ***four*** questions selecting one from each unit. These will be in addition to the compulsory question at serial number I and each question will carry 15 marks.

3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.

BOOKS RECOMMENDED

Essential Readings

1. Berry, B.J.L.: Geography of Market Centres and Retail Distribution, Englewood Cliffs, Prentice Hall, 1966.

2. Breese, G.: Urbanisation in Newly Developing Countries, Prentice Hall, New Delhi, 1969.

3. Gibbs, J. P. (ed): Urban Research Methods, Affiliated East-West Press, New Delhi, 1966.

Further Readings

1. Haggett, P.: Locational Analysis in Human Geography, Arnold, London, 1965.

2. Meadows,P. and Hizruchi,E.H. (eds): Urbanism, Urbanisation and Change, Addoson-Wesley Publishing Company, London, 1969.

Opt. (viii): An Advanced Course in Geography and Ecosystems

Max. Marks	: 100
Terminal Examination	: 80
Internal Assessment	: 20
Time	: 3 Hours

Objectives:

(i) To appraise the students with the interrelationship between humans and the environmental and the linkages with other organizations.

(ii) To generate awareness about bio diversity conservation and its importance in maintaining ecological balance.

(iii) Examples of human induced ecological changes will be given and restoration measures suggested.

UNIT I

i. Conceptual basis: nature and wilderness, ecology and environment, ecosystem stability and resilience.

UNIT II

ii. Recent developments in the field and emerging issues.
iii. Sources of data and research methods including use of remote sensing and CIS.

UNIT III

iv. Geographical perspective on ecosystems: biodiversity, hot spots, island biogcogniphy, clinitite change, carrying capacity, landuse/landcover change.

UNIT IV

v. Environment and development with special focus on environmental degradation, sustainable development.

vi. Case studies on: Ecological problems in North West India with special reference to: deforestation, flooding, landslides. Green Revolution and wetlands.

Note:

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. The student shall attempt any 10 parts in about 25-30 words each. Each part shall carry 2 marks (total 20 marks).

2. A total of ***eight*** questions will be set out of the whole syllabus, at least *two* from each unit. The candidates will attempt ***four*** questions selecting one from each unit. These will be in addition to the compulsory question at serial number I and each question will carry 15 marks.

3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.

Pedagogy:

Interaction between teachers and students on different aspects of ecology with the help of models, charts and pictures. Emphasis on the environmental problems faced by India in recent years with an emphasis on India

Books recommended:

Essential Readings

1. Alexander, Roy, and Andrew C. Millington, 2000. **Vegetation Mapping: From Patch to Planet.** Chichcstcr: John Wiley and Sons.
2. Burroughs, William James, 2001: **Climate Change: A Multidisciplinary Approach**. Cambridge: Cambridge University Press.
3. Ernst, W. G., (cd.) 2000: **Earth Systems: Processes and Issues**. Cambridge: Cambridge University Press.
4. Glantz., Michael H., 1996: **Currents of change: El Ninons impact on Climate and Society.** Cambridge: Cambridge University Press.
5. Johnston, R.J., Peter J. Taylor and Michael J. Watts, (eds.) 2002: **Geographies of Global Change: Remapping the World**. Oxford: Blackwell Publishing.

6. Kirkby, John, Phil O'Keefe, and Lloyd Timberlake, 1995. **The Earthscan Reader in Sustainable Development.** London: Earth scan Publications.

7. Mawhinney, Mark, 2002: **Sustainable Development: Understanding the Green Debates**. Oxford: Blackwell Science.

8. Millington, Andrew C., Stephen J. Walsh. and Patrick E. Osborne, 2001: **GIS and Remote Sensing Applications in Biogeography and Ecology.** Boston: Kluwer Academic Publishers.

9. Osborne, Patrick L., 2000: **Tropical Ecosystems and Ecological Concepts**. Cambridge: Cambridge University Press.

10. Rcid, Dtivid, 1995: **Sustainahic Development: An Introductory Guide**. London: Earthscan Publications.

11. IUCN, UNEP, and WWF, 1991: **Caring for the Earth: A Strategy for Sustainable Living.** Gland, Switzerland; Published in partnership by 1UCN- The World Conservation Union, UNEP- United Nations Environment Program and WWF- World Wide Fund for Nature.

Further readings:

1. Faeth, Paul, (cd.) 1993: **Agricultural Policy and Sustainability**: Case Studies from India, Chile, the Philippines and (the United States. Washington: World Resources Institute.

2. Holloway, Sarah L., Stephen P. Rice, and Gill Valentine, (eds.) 2003: **Key Concepts in Geography.** London: Sage Publications.

3. Indian Institute of Remote Sensing, 2002: **Biodiversity Characterization at Landscape Level in Western Himalayas, India using Satellite Remote Sensing and Geographic Information System.** Dehra Dun: Indian Institute of Remote Sensing.

4. Williams, Martin A. J. and Robert C. Balling Jr, 1996: **Interactions of Desertification and Cliimate** London: Arnold.

C. Dissertation (6 Credits): 150 Marks

Each candidate will be required to write a dissertation on a subject approved by the Board of Control. He/she will be required to submit the dissertation by the end of the second semester. In case, a candidate is unable to complete and submit the dissertation in time, he/she will have to seek and obtain an extension in time, as per regulations.

SECOND SEMESTER

Course IV: Geography of Regional Development

Max. Marks : 100
Terminal Examination : 80
Internal Assessment : 20
Time : 3 Hours

Objectives: The course aims at exposing the students conceptual and methodological issues involved to understand regional development process and causes and consequences to widening regional inequalities in development. The basic focus will be placed on analytical examination of historical developments in development process and various modals and theories propounded by different scholars to explain spatial dimensions of economic growth along with the study of policies and programs adopted in India to remove regional inequalities in development.

UNIT I

1. Conceptual framework: definition, development studies and social sciences, development and geography, economic development and regional development, regional disparity and regional diversity, issues in regional development planning

UNIT II

2. Development models: descriptive, analytical and normative aspects of the development process, stages of growth models, spatial models

3. Theories related to spatial dimensions of economic growth by G. Myrdal and A.O. Hirschman; regional growth and location theories by E. M. Hoover, W. Isard and A. Losch; growth pole theories by F. Perroux and A. Kuklinski ; theories of spatial integration and spatial organization by J. Friedmann and E.A.J. Johnson; theories of regional underdevelopment by A.G. Frank and D. Slater; theories of decentralized development by D. A. Rondinelli and C. Weaver; and economic growth and regional development theories of B. Ohlin, D.C. North and H.S. Perloff.

UNIT III

4. Measurement of regional disparities; issue of the spatial scale; economic, social, political, infrastructural and ecological indicators of development, techniques of identifying regional disparities

5. Strategies of regional development and planning: integrated regional development, basic needs approach, human development, target area and target group approach, balanced versus unbalanced approach, wage goods versus heavy industry approach, strategies for special problem regions (hill, tribal, desert, drought prone, coastal, border, naxal affected and backward districts initiatives)

UNIT IV

6. Regional development planning exercises in India-the South-East Resource Region Plan, the Western Ghats Plan, the Rayalseema Plan, National Capital Region Plan, Indira Gandhi Canal Command Area Plan

7. Economic reforms and regional development in India, Regional disparities in India

Note:

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. The student shall attempt any 10 parts in about 25-30 words each. Each part shall carry 2 marks (total 20 marks).

2. A total of ***eight*** questions will be set out of the whole syllabus, at least *two* from each unit. The candidates will attempt ***four*** questions selecting one from each unit. These will be in addition to the compulsory question at serial number I and each question will carry 15 marks.

3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc

Books Recommended

Essential Readings

1. Anthony R.de Souza and J.Brady Foust : *World Space Economy*, A Bell and Howell Company, Columbus.

2. A. Kundu and M. Raza: *Regional Dimension of Indian Economy*, Spektrum, New Delhi, 1982.

3. Charles Gore: *Regions in Question*, Methuen, London, 1984.

4. E. A.J.Johnson: *The Organisation of Space in Developing Countries*, Harvard University Press, Cambridge, 1970.

5. J. Friedmann and W. Alonso: Regional Policy: Readings in Theory and Applications, M.I.T. Press, Cambridge, 1975.

Further Readings

1. J. Friedmann and W.C. Weaver: *Territory and Function*, Arnold, London, 1979.

2. John Friedmann and William Alonso: *Regional Development and Planning*, M.I.T. Press, Cambridge.

3. Nina Singh: *Administration and Development of Indian States*, Anmol Publications, New Delhi, 1998.

4. R.J. Chorley and Peter Haggett (eds): *Models in Geography*, Methuen, London, 1976.

5. R.P. Misra and Others: *Regional Development Planning in India*, Vikas, New Delhi, 1974.

6. Stuart Corbridge (ed): *Development Studies: A Reader*, Arnold, London, 1995.

Course V: Digital Cartography Map Design and Remote Sensing

Max. Marks : 100
Terminal Examination : 80
Internal Assessment : 20
Time : 3 Hours

Objectives:

The objective of this course is to introduce the students about various concepts in digital cartography, GIS and Remote Sensing. They are also taught different concepts about map design and layout. They are also told about benefits of GIS and Computer-assisted cartography.

Unit I

(i) Nature, Scope and Concepts in Digital Cartography: Cartographic Symbolization; Communication and Perception: Map generalization; Map design and layout.

Unit II

(ii) Spatial Data base and Data Models: Points, Lines and Polygons; Raster and Vector Data models.

(iii) Designing of Maps, Drafting techniques, Drafting materials: Base map compilation for thematic mapping; planning of thematic map series and thematic atlases.

Unit III

(iv) Map Reproduction: Reproduction of maps through various printing and non-printing processes.

Unit IV

(v) Introduction to Environmental Remote Sensing: Definition, historical development, types of remote sensing and sensors, remote sensing platforms and sensor packages; Interpretation of remotely sensed data.

Note:

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. The student shall attempt any 10 parts in about 25-30 words each. Each part shall carry 2 marks (total 20 marks).
2. A total of ***eight*** questions will be set out of the whole syllabus, at least *two* from each unit. The candidates will attempt ***four*** questions selecting one from each unit. These will be in addition to the compulsory question at serial number I and each question will carry 15 marks.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc

Books Recommended:

Essential Readings

1. Brown, L.A.: The story of Maps, Dover Publications, Inc. New York.

2. Monkhouse, F.J. and Wilkanson. H.R.: Maps and Diagrams, 3rd ed, Methuen, London, 1971.

3. Robinson, A.H. and Sale. R.D.: Elements of Cartography, John Wiley and Sons, New York (latest edition).

4. Sabins, JR. P.P.: Remote Sensing: Principles and Interpretation, W.H. Freeman and Company, New York. 1986.

Further Readings:

1. Bagrew.L.: *History of Cartography*, C.A. Watts and Co., London, 1964.

2. Barren, E.C. and Curtis, L.F.: *Introduction to Environmental Remote Sensing*, Chapman and Hall Ltd., London, 1976.

3. Cuff, D.J, and Mattson, M.T.: Thematic Maps, Methuen, New York, 1982.

4. Deekshatulu. B.L. and Rajan, Y.S. (eds.): Remote Sensing, Indian Academy of Sciences, Bangalore, 1984.

5. Keates, J.S.: Understanding Maps, Longman, London, 1982.

6. Keates, J.S.: Cartographic Design and Production, Longman, London, 1978.

7. Lawrence, G.R.P.: Cartographic Methods, Methuen, London, 1971.

8. Cromley, R.G: Digital Cartography, Englewood Cliff, Prentice Hall. 1992.

Pedagogy:

There should be adequate interaction between the teacher and students. The teacher should make maximum use of wall maps and other illustrations like maps from geography books, Ph.D. thesis, research reports and atlases while teaching the use of different cartographic techniques. This course is concerned with visual techniques therefore maximum use of the visual illustrations should be made while teaching this course.

Dissertation

The candidate will be required to complete and submit the dissertation on the approved topic by the end of the semester.

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